GRADE 2 · DEBUGGING & READING CODE

Read Two Scripts Carefully

Trace both, move by move

Name: .... Date: ....

★ **I can read two scripts that run at the same time and find where each one lands.**

Good coders read carefully

Before you can fix code, you must READ it carefully. Bit (green) and Pixel (purple) both start on the green flag, so they run at the same time. Each one has TWO moves. Read each script one move at a time and find where each sprite lands.

The number path

0	1	2	3	4	5
Bit					Pixel

Bit

Bit's script

⚑ when green flag clicked

forward 2

forward 1

Pixel

Pixel's script

⚑ when green flag clicked

back 1

back 1

1 Read Bit's script

Trace Bit from home 0: forward 2, then forward 1. What number does Bit land on?

2 Read Pixel's script

Trace Pixel from home 5: back 1, then back 1. What number does Pixel land on?

3 Put them together

Both start on the same green flag. Write where each lands. Bit: ____ Pixel: ____. Do they land on the same number?

Big idea

Reading code means tracing each block in order. With two scripts running at once, read each one carefully on its own, then put the results together. Careful reading is the first step to finding bugs.

TEACHER GUIDE · GRADE 2 · DEBUGGING & READING CODE

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Quick facilitation notes & answer key

What this worksheet teaches

The first debugging sheet builds the key skill: reading two sets of moves accurately. Both sets are correct here (no bug yet) — students trace each one move by move and confirm where the two characters land. Careful reading comes before bug-finding. No coding experience needed.

Before you start (no computer needed)

No computer needed — paper and a pencil. You read the prompts aloud. Optional: two counters on a number line 0 to 5 (green Bit, purple Pixel) so children can trace both sets of moves and see which one misses the goal.

How to lead it (about 20 minutes)

1. Show them (you do). Trace Bit one move at a time, then Pixel one move at a time.
2. Do one together. Exercise 1 — Bit lands on 3; Exercise 2 — Pixel lands on 3.
3. Let them try. Students write both landings and check if they meet (Ex 3).

Answer key (these are the verified correct answers)

Exercise 1 — Bit: 0, forward 2 → 2, forward 1 → 3. Exercise 2 — Pixel: 5, back 1 → 4, back 1 → 3.

Exercise 3 — both start on the green flag; Bit lands on 3 and Pixel on 3 → same number, they meet.

Note: there is no bug on this sheet — the point is accurate tracing, which the next sheets build on to find bugs.

If students get stuck — and ways to adjust

Watch for: mixing the two characters' moves. Finish one fully, then the other.

Make it easier: trace Bit with a green counter, then Pixel with a purple counter.

Make it harder: ask how they would check a trace is right (re-run it slowly).

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

In plain terms: students write and run instructions, including two sets that run at the same time.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

In plain terms: students read two sets of instructions, find and fix the wrong one, and explain the fix.

You'll know it worked when

The child traces both sets accurately and confirms they meet on 3.